

Parent Guide



@twinklparents

We're excited to share this activity with you. If you are interested in finding more engaging, fun and interesting activities for you and your children, then check out these links to different areas of the [Twinkl Parents](#) website.

games



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word searches



What is this resource and how do I use it?

If your child is learning about algebra in their maths lessons, these activities could be the perfect accompaniment. Alternatively, use them as a way of recapping everything your child has learnt before any upcoming tests. There are explanations, examples and questions about formulae, patterns, missing number problems and two unknowns.

What skills does this practise?

Simple Formulae

Patterns and Sequences

Missing Number Problems

Algebra - Two Unknowns

Further Activity Ideas and Suggestions

Keen to explore algebra in more detail? Take a look at [this category](#). Your child can check they know the algebra [shorthand rules](#) or can practise [using brackets](#). Why not play this [board game](#) for some fun algebra revision?

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Algebra Simple Formulae

The basic idea behind algebra is to replace a number with a letter or symbol.

For example, we know that $5 \times 2 = 10$.

If we were presented with $5 \times ? = 10$, we might know straight away that the missing number is 2. The process we do automatically is $10 \div 5 = 2$, so $? = 2$.

We might be presented with something that appears to be trickier.

For example, $138 + z = 314$

z stands for the unknown number.

It could be a circle or a leaf or a spiral or a cake... it doesn't matter.
It's just there to show that there is an unknown quantity that we need to find.

To solve this, we need to get z on its own. To do that, we can see that we could take away 138. We must do the same to both sides of the equation.

The diagram shows two boxes. The top box contains the equation $138 + z = 314$. The bottom box contains the equation $z = 314 - 138$. Two red curved arrows point from the top box to the bottom box. The left arrow starts at the 138 in the top equation and points to the -138 in the bottom equation. The right arrow starts at the 314 in the top equation and points to the 314 in the bottom equation.

You can use any method you like to work out this subtraction.

You should be left with...

$$z = 176$$

You can check by putting this value back into the equation.

$$138 + 176 = 314$$

$$314 = 314$$

It works; both sides are balanced.

Algebra Simple Formulae

Letters and Numbers

You might have seen something like this:

$$5a + 2 = 17$$

What does 5a mean?

When we write simple formulae like this, 5a means 5 x a. It indicates that the two amounts are being multiplied together.

So, to solve this one, we would first take away 2 from both sides of the equation...

$$5a + 2 = 17$$

$$5a = 17 - 2$$

$$5a = 15$$

Then, to get a on its own, we need to divide it by 5. We have to do the same for the other side.

$$a = 3$$

You can check it by putting this value back into the equation.

$$(5 \times 3) + 2 = 17$$

$$15 + 2 = 17$$

$$17 = 17$$

Both sides are balanced!

Algebra Simple Formulae

Solving Simple Formulae

Can you solve these algebra problems? Remember, you need to get the unknown on its own. Always do the same to both sides of the equation.

a $20 = 4a + 4$

e $2e - 5 = 5$

b $3b + 5 = 11$

f $21 = 3f + 3$

c $14 = 6c - 4$

If $g = 6$, can you work out what h is?

$h = 5g + 14$

d $48 = 12d$

Algebra: Number Sequences and Patterns

Algebra can be very useful when explaining patterns or sequences.

For example, let's look at this sequence of numbers:

1 st term	2 nd term	3 rd term	4 th term
8	11	14	17

We are given the formula $3n + 5$ for this sequence. We would multiply the term value by 3 then add 5. This means that we can use this formula to calculate any number in the sequence.

Without using the formula, we can see that this sequence goes up by 3 each time. Therefore, we can guess that the next number in this sequence (the 5th term) will be 20.

Let's use the formula to check. We are finding the 5th term, so $n = 5$.

$3 \times 5 + 5$	$\rightarrow$	$15 + 5 = 20$
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It works!

This formula is useful when finding a distant term. For example, we might be asked to find the 19th number in this sequence. Instead of working out all the other numbers in between, we can put $n = 19$ straight into our formula.

$3 \times 19 + 5$	$\rightarrow$	$57 + 5 = 62$
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Therefore, the 19th term = 62.

Algebra: Number Sequences and Patterns

Number Sequences

Use your understanding of number sequences to answer these questions.

a The formula $4n + 1$ can be used to calculate the value of the terms in this sequence:

5 9 13 17 21

Fill in the missing information in this table:

Term	Calculation	Value
1 st	$4 \times 1 + 1$	5
6 th		
10 th		
20 th	$4 \times 20 + 1$	

b The formula $3n - 1$ can be used to calculate the value of the terms in this sequence:

2 5 8 11 14

Fill in the missing information in this table:

Term	Calculation	Value
1 st	$3 \times 1 - 1$	2
7 th		
12 th		
18 th	$3 \times 18 - 1$	

Algebra: Number Sequences and Patterns

- c** 11 is the third term in this sequence. Circle the formula that could be used to calculate this term.

3 7 11 15 19

$3 \times 4 - 1$

$3 \times 5 - 1$

$3 \times 4 + 1$

- d** 17 is the fourth term in this sequence. Circle the formula that could be used to calculate this term.

2 7 12 17 22

$4 \times 4 + 2$

$4 \times 5 - 3$

$4 \times 5 - 2$

Algebra: Missing Number Problems

Algebraic equations can be used to express missing number problems.

Age in an Equation

We could use one to show the difference in age between two siblings. For example, Ben and Eli were born on the same day but 2 years apart.

Ben is 2 years older. We could write this as an equation:

$$e + 2 = b$$

If we add 2 to Eli's age, we can work out Ben's age.

When Ben is 31, how old will Eli be?

$$e + 2 = b$$

$$e + 2 = 31$$

$$e = 31 - 2$$

$$e = 29$$

Eli will be 29.

When Eli is 17, how old will Ben be?

$$e + 2 = b$$

$$17 + 2 = b$$

$$b = 19$$

Ben will be 19.

Algebra: Missing Number Problems

Decorator Costs Equation

Equations can be useful for calculating costs. For example, if a decorator charges £12 per hour, plus £50 for materials, how could we represent this as an equation?

Let's use h for hours.

If the decorator charges £12 per hour, we can write 12h to show that the number of hours must be multiplied by £12.

The decorator adds on £50 for materials. We can show this by writing + 50 in the equation.

$$\underline{12h} + 50$$

If the decorator worked 11 hours on a project, what would the cost be?

Let's use the equation...

$$12h + 50$$

$$12 \times 11 + 50$$

$$132 + 50$$

£182 total cost

Algebra: Missing Number Problems

Missing Number Equations

1a A plumber charges £14 per hour and adds on £75 for materials. Can you write an equation to show how to calculate the total cost? Use ***h*** for hours.

1b How much would you pay the plumber for 15 hours of work?

2a A hairdresser charges £12 per hour but is offering a £5 discount. Write an equation to show this, using ***h*** for hours.

2b How much would you pay for a 2-hour session?

Algebra: Missing Number Problems

- 3a** Floor tiles are £24 per square metre. You also need to pay £40 for someone to come and lay the tiles. Write this as an equation, using **m** for square metre.

- 3b** Bhavesh's kitchen is 5 square metres. How much will he pay for a new tiled floor?

Parent Tip: It helps to highlight the key information in each question, then test out the formula to check if it makes sense.

Algebra: Two Unknowns

Sometimes, there might be two unknowns in an equation. It could be that you have to find all the possible quantities that could fit into the equation.

For example, $ab = 18$.
This means $a \times b = 18$.

We only know that a and b are both whole numbers.
What could these values be?

In this situation, we need to think of pairs of numbers that, when multiplied together, make 18.

Working systematically,
we could start with:

$$1 \times 18$$

$$2 \times 9$$

$$3 \times 6$$

After this, we would be swapping
the values of a and b around.

$$18 \times 1$$

$$9 \times 2$$

$$6 \times 3$$

These are all the possibilities if a and b are whole numbers.

Algebra: Two Unknowns

Sometimes, you might be given a few pieces of information about the unknown values. a and b are, once again, whole numbers.

$$a < 5$$

$$2a + b = 10$$

$$b < a$$

From this, we know that a could be 1, 2, 3 or 4.

We also know that b is less than a .

Therefore, a cannot be 1 because there is no whole number less than 1.

Let's test a possible value.

If a was 2...

$$2 \times 2 + b = 10$$

$$4 + b = 10$$

$$b = 10 - 4$$

$$b = 6$$

We know that this can't be right because $b < a$ and 6 is not less than 2.

If a was 3...

$$2 \times 3 + b = 10$$

$$6 + b = 10$$

$$b = 10 - 6$$

$$b = 4$$

We know that this can't be right because $b < a$ and 4 is not less than 3.

If a was 4...

$$2 \times 4 + b = 10$$

$$8 + b = 10$$

$$b = 10 - 8$$

$$b = 2$$

This works! 2 is less than 3.

Algebra: Two Unknowns

Two Unknowns

- a** Find three possible pairs of numbers for **a** and **b** in this equation (**a** and **b** are whole numbers):

$$19 = ab + 7$$

Value of a	Value of b

- b** Find three possible pairs of numbers for **c** and **d** in this equation (**c** and **d** are whole numbers):

$$13 = cd - 7$$

Value of a	Value of b

Algebra: Two Unknowns

c Find the value of **e** and **f**.

$$ef = 21$$

$$e + f = 10$$

$$e < f$$

$$e =$$

$$f =$$

d Find the value of **g** and **h**.

$$g - h = 3$$

$$g + h = 9$$

$$g =$$

$$h =$$

Parent Tip: It can help to have some scrap paper or a mini whiteboard on hand for scribbling down ideas and testing values in the equation.

Algebra: Solving Simple Formulae Answers

- a) $a = 4$ If $g = 6$, can you work out what h is?
b) $b = 2$ $h = 5g + 14$
c) $c = 3$ $h = 30 + 14$
d) $d = 4$ $h = 44$
e) $e = 5$
f) $f = 6$

Algebra: Number Sequences Answers

a)

Term	Calculation	Value
1 st	$4 \times 1 + 1$	5
6 th	$4 \times 6 + 1$	25
10 th	$4 \times 10 + 1$	41
20 th	$4 \times 20 + 1$	81

b)

Term	Calculation	Value
1 st	$3 \times 1 - 1$	2
7 th	$3 \times 7 - 1$	20
12 th	$3 \times 12 - 1$	35
18 th	$3 \times 18 - 1$	53

- c) $3 \times 4 - 1$
d) $4 \times 5 - 3$

Algebra: Missing Number Problems Answers

- 1a) $14h + 75$ 2a) $12h - 5$ 3a) $24m + 40$
1b) £285 2b) £19 3b) £160

Algebra: Two Unknowns Answers

a) Any three of these possibilities:

Value of a	Value of b
1	12
2	6
3	4
4	3
6	2
12	1

b) Any three of these possibilities:

Value of a	Value of b
1	10
2	10
4	5
5	4
10	2
20	1

c) $e = 3$ $f = 7$

d) $g = 6$ $h = 3$